8.2 Answers: ratio and scale

← Unit overview · Practice

For the parent: if he has the right table but an arithmetic slip, say so first (“Your table is exactly right”), then look at the slip together.

Set A: Ratio tables

1. **12 yoghurts.** 3 × 4.
2. **€6.** Double the kilos, double the price.
3. **€4 and €6.** 10 pens = 2 × 5 pens; 15 pens = 3 × 5 pens.
4. **Table:**

personen	4	8	2	6
bloem (g)	250	500	125	375
melk (ml)	500	1000	250	750
eieren	2	4	1	3

   For 6: add the column for 4 and the column for 2.
5. **€1,50 and €10,50.** Divide by 3 for one; then 7 × €1,50.
6. **150 ml water, 200 ml lemonade.** 3 × 50 ml water; 50 + 150 in total.
7. **12 children.** 30 is 6 groups of 5; 6 × 2.
8. **20 sweets.** 2 × 10.

Set B: Scale on maps and plans

1. **5 m.** 1 cm = 100 cm = 1 m.
2. **30 m.** 1 cm = 1 000 cm = 10 m; 3 × 10 m.
3. **250 m; 4 cm = 1 000 m = 1 km.** 25 000 cm = 250 m.
4. **2,5 km.** 10 × 250 m = 2 500 m.
5. **3 km.** 1 cm = 500 m; 6 × 500 m = 3 000 m.
6. **7,5 km.** 1 cm = 100 000 cm = 1 km.
7. **8 cm.** 1 cm on the drawing = 50 cm real; 400 cm : 50.
8. **8 cm.** 4 cm = 1 km, so 2 km = 8 cm.
9. **1 : 50 000.** 1 cm = 500 m = 50 000 cm.
10. **18 cm.** 1,80 m = 180 cm; 180 : 10.

Set C: Speed in km/u

1. **120 km.** 2 × 60.
2. **15 km; 7,5 km.** Halve both for half an hour.
3. **15 km.** 3 × 5.
4. **60 km.** Half an hour → half the distance.
5. **2 hours 30 minutes.** 250 km is 2,5 × 100 km; 0,5 hour = 30 minutes.
6. **24 km/u.** 30 min → 60 min is ×2, so 12 km × 2.
7. **6 km/u.** 20 min × 3 = 60 min, so 2 km × 3.
8. **80 km/u.** Double both: 80 km in 60 minutes.
9. **6 km.** 60 min : 3 = 20 min, so 18 km : 3.
10. **60 km.** 45 km in 60 min; 15 km in 20 min; together 60 km in 80 min.

Set D: Recipes and the best buy

1. **The 2-litre pack.** €1,20 per litre vs €2,00 : 2 = €1,00 per litre.
2. **The box of 10.** €2,40 : 6 = €0,40 per egg; €3,50 : 10 = €0,35 per egg.
3. **The 300 g bag.** €0,75 per 100 g vs €2,10 : 3 = €0,70 per 100 g.
4. **The 600 g pack.** €2,40 : 4 = €0,60 per 100 g; €3,30 : 6 = €0,55 per 100 g.
5. **€9 for two; €4,50 per bottle.** €6 + €3.
6. **450 g.** For 2: 150 g; for 6: 300 g + 150 g.
7. **225 g.** For 10 cookies: 75 g; for 30: 150 g + 75 g.
8. **200 ml syrup and 800 ml water.** 1 + 4 = 5 parts; 1 000 ml : 5 = 200 ml per part.
9. **The 5 kg bag.** €4,50 : 5 = €0,90 per kg; €2,00 : 2 = €1,00 per kg. (But only worth it if you will eat 5 kg before they spoil.)
10. **10 boys and 15 girls.** 2 + 3 = 5 parts; 25 : 5 = 5 per part; boys 2 × 5, girls 3 × 5.
11. **2 eggs.** Twice as many people, twice as many eggs.

Puzzle corner

1. **About 8 1/3 days (8 days and 8 hours).** 1 km = 1 000 m; 1 000 : 5 = 200 hours; 200 : 24 = 8 remainder 8.
2. **Each side is 500 m.** 2 cm × 250 m. Map area 4 cm², real area 500 m × 500 m = 250 000 m². In cm² that is 25 000 × 25 000 = 625 000 000 times bigger: the scale applies in both directions.
3. **After 3 hours.** The first candle has ¼ left (3 of its 4 hours used), the second has ½ left (3 of its 6 hours used). ¼ is half of ½.